ZCAM LATAM HUB / SYSTEM ADMINISTRATOR

Azure Databricks Platform - Target Architecture + Terraform

Hub-and-Spoke | One reusable Terraform module | Separate DEV / PROD subscriptions | Private data access

Controlled API egress

DATA ENGINEERING TEAM / 15
São Paulo office
Corporate VPN access
Entra ID / conditional access

EXISTING HUB VNET — Shared network services · CIDR to be confirmed

- **VPN GATEWAY**: Existing corporate entry; São Paulo office
- **AZURE FIREWALL**: FQDN-based HTTPS egress; Approved API destinations
- **PRIVATE DNS**: Resolver + zone links; blob / dfs / sql / kv / adb
- **PRIVATE ACCESS**: Workspace UI / SSO; Private endpoints + auth

EXTERNAL APIs × 2
HTTPS 443 only
FQDN allowlisting
Non-PII ingestion

Hub / Spoke peering

Hub / Spoke peering

PRODUCTION SUBSCRIPTION / Spoke VNet 10.20.0.0/16

DATABRICKS PREMIUM [TF] — adb-zcam-prod-brs · Secure Cluster Connectivity

- **Host + container subnets**: snet-adb-host-prod / snet-adb-container-prod; Delegated + NSGs; no public IP
- **PRIVATE LINKS / OUTSIDE TF**: ADB UI / backend · ADLS blob + dfs; SQL · Key Vault · browser auth · DNS

DATA SERVICES / PUBLIC NETWORK ACCESS DISABLED — API pull via Hub Firewall

stlatamraw → **stlatamcurated** → **stlatamanalytics**

- stlatamraw: Non-PII · approved HTTPS pull
- stlatamcurated: Customer PII
- stlatamanalytics: Aggregated BI-ready

- **PRODUCTION KEY VAULT**: Ingestion reads API secrets · rotation + PE
- **UNITY CATALOG + ENTRA ID**: Least privilege · auditing · PII grants

DEVELOPMENT SUBSCRIPTION / Spoke VNet 10.30.0.0/16

DATABRICKS PREMIUM [TF] — adb-zcam-dev-brs · Secure Cluster Connectivity

- **Host + container subnets**: snet-adb-host-dev / snet-adb-container-dev; Delegated + NSGs; no public IP
- **PRIVATE LINKS / OUTSIDE TF**: ADB UI / backend · ADLS blob + dfs; Key Vault · browser auth · DNS

DEV DATA SERVICES / ISOLATED

Raw / Curated / Analytics storage

Synthetic or irreversibly de-identified data only

- **DEVELOPMENT KEY VAULT**: Dev identity reads secrets · own rotation
- **DEV ENTRA GROUPS + JOB POLICY**: Auto-termination · no production PII

SQL over Private Link

DR replication / recovery

EXISTING AZURE SQL DATABASE [EXTERNAL]
Hosted in another subscription · Production Private Endpoint
Entra authentication; DNS private resolution
Recovery and permissions require SQL owner approval

PRODUCTION DISASTER RECOVERY [DESIGN]
Secondary approved region · Warm private networking foundation
Data replication + redeployable workspace and jobs
RPO target: 4h | RTO target: 2h | Validate with recovery drills

TERRAFORM - ONE REUSABLE MODULE

Workspace scope only · Dependencies not provisioned

PRODUCTION ROOT
environments/prod/main.tf
environment = prod | Prod subscription
Independent Terraform state (remote backend planned)
Vars: subscription / RG / VNet / Access Connector

calls

SHARED CHILD MODULE / CREATED RESOURCE
modules/databricks/main.tf
azurerm_databricks_workspace.this
Premium · VNet injection · no_public_ip · private
Existing delegated subnets + NSGs + connector required

calls

DEVELOPMENT ROOT
environments/dev/main.tf
environment = dev | Dev subscription
Independent Terraform state (remote backend planned)
Vars: subscription / RG / VNet / Access Connector

OUTSIDE TERRAFORM v2: Hub / Spokes / subnets / NSGs / Private Endpoints / DNS / ADLS / SQL / Key Vault / UC grants / DR

KEY: Blue - private network | Green - protected data / DR | Gold - HTTPS egress | Purple - Terraform scope

Est. USD 7,567/mo < USD 8k | RPO 4h / RTO 2h: targets only | Not deployed